

An aerial photograph of Halifax, Nova Scotia, Canada. The image shows a dense urban area with numerous houses and buildings. A large body of water, the harbor, is visible in the center, with several sailboats and larger vessels. The surrounding landscape is a mix of green trees and brown autumn foliage. The sky is clear and blue.

Sustaining Halifax's Urban Forest

Urban Forest Master Plan (UFMP)

June 13, 2016

Process of plan development

- **Regional Council motion –Aug. 28, 2001**
- **Regional Plan urban forest research and policy-2001-2006**
- **Urban forest inventory – 2007-10**
- **Early public engagement – 2010 – values**
- **Analysis and writing – 2011**
- **Second-round engagement – 2011-12**
- **Revisions – winter 2012**
- **Third-round engagement – May–July 2012**
- **Implementation strategy- UFMP Chapter 7 - Aug. 2012**
- **HRM Environment and Sustainability Standing Committee (ESSC) endorsement of UFMP-Sept. 6,2012**
- **ESSC report to Regional Council-Sept. 25,2012**
- **UFMP adopted by Regional Council –Sept. 25, 2012**

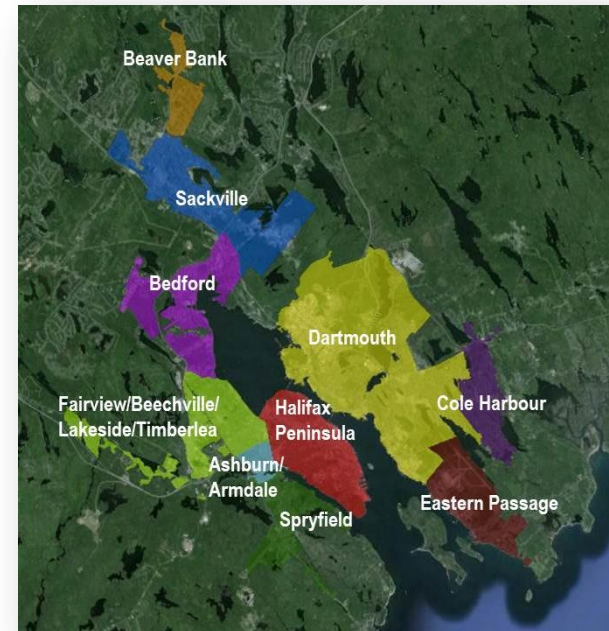


A photograph of a paved path in a park, lined with large, mature trees. The path leads into the distance where a group of people is walking. On the left side of the path, there are green benches and a black trash can. The trees have green leaves, and the grass is green. The scene is bright and sunny.

Fact: people of HRM say that trees in the city are very important to them.

What is an urban forest?

- Every tree in HRM's serviced communities
- Includes trees on private as well as public property
- 709,000 publicly-owned trees on HRM roads and streets
- 157,000 directly planted & managed
- 552,000 naturally regenerated
- 94,000 plantable spots for trees on HRM rights of way



Why is the urban forest important?

- Urban forests are a bio-utility and a major part of every city and town's infrastructure
- Trees are a critical aspect of urban landscapes that must be managed sustainably
- Trees deliver valuable environmental, social, health and economic benefits and services





air quality benefits – Urban trees play a large role in reducing airborne pollutants by exchanging gases with the atmosphere and capturing chemical pollutants and particulates that can be harmful to people



About 1,478 metric tons (Mg) of pollutants are removed annually by urban trees and shrubs in the serviced area of HRM.

This equates to \$9.6 million each year in air pollution mitigation benefits.

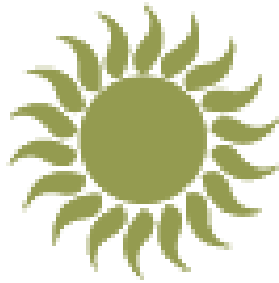


biodiversity – Increasing the populations of native trees in the urban setting gives the opportunity for native birds, small mammals, insects and other native species to occupy habitat niches in the city.



Flowers, fruits, leaves, buds and woody parts of trees are used by many different species.

Bacteria and fungi contained in tree parts cause decay which makes nesting easier for some birds (Urban Forestry Network, 2014).



carbon sequestration benefits –

By constantly capturing and accumulating carbon-based greenhouse gases in their roots, stems and branches, trees can help mitigate the effects of climate change.



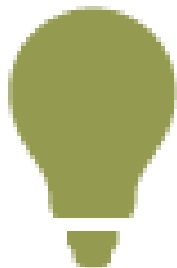
HRM's street trees are estimated to hold more than 2 million tonnes of carbon, and annually sequester more than 100,000 tonnes.



economic benefits – Street trees and private trees add significantly to the overall value of residential properties.



The presence of larger trees in yards and as street trees can add from 3% to 15% to home values throughout neighborhoods (Wolf, 2007).



energy use benefits – Shade provided by urban trees can reduce the total energy required to cool a building.



HRM's street trees save \$12.4 million in energy costs each year.



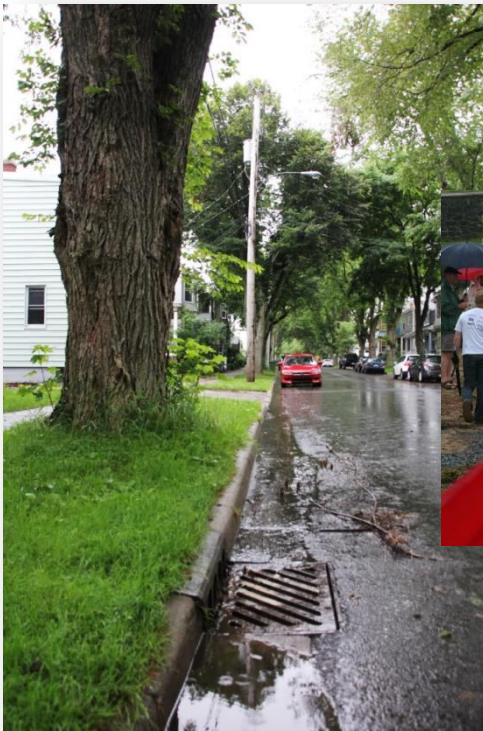
health benefits – Trees improve the quality of air we breathe, and their presence has been shown to reduce stress.



Hospital patients with views of trees heal significantly faster and with fewer complications than those without such favourable views (Ulrich, 1984).



hydrological benefits – Urban forests play an essential role in moderating stormwater and flood damage, water quality, erosion, and stormwater treatment costs.



HRM's street trees provide about \$2.1 million in stormwater reduction services annually.



noise-reduction benefits – Urban trees are effective noise buffers, which is a major asset for residential neighbourhoods near highways or industrial areas.



A row of conifers can reduce traffic noise by 50% or more (Reethof & McDaniel, 1978).



road benefits – Trees can make our roads safer. A tree-lined street gives the illusion that the street is narrower, and generally calms traffic.



A road surface that is 20% shaded lasts longer. After 30 years, this results in a 60% savings on resurfacing (McPherson *et al.*, 1999).



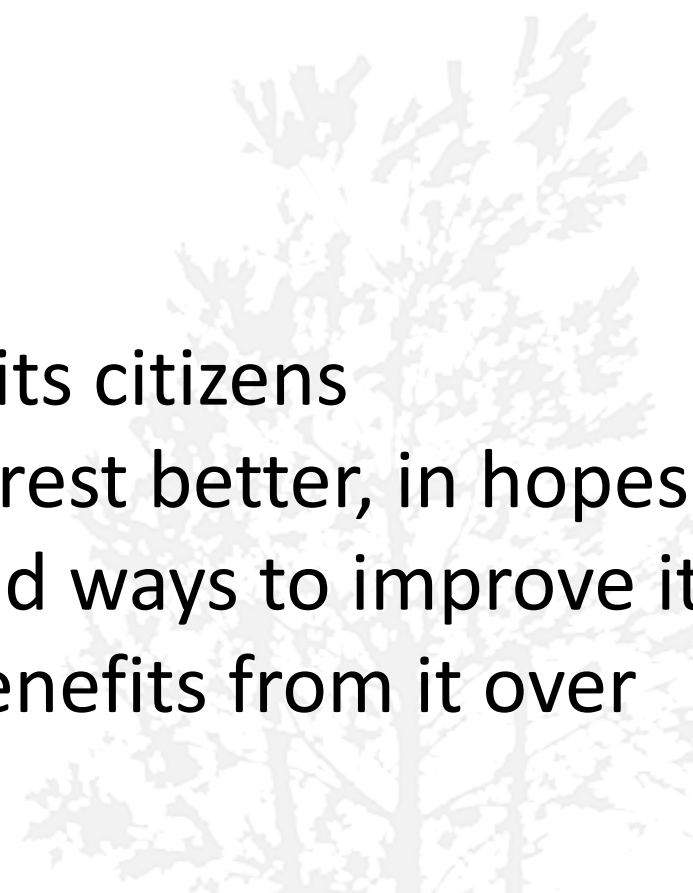
social & psychological benefits –

People like trees around them because they make life more enjoyable and urban conditions more tolerable.



The experience of nature helps to restore the mind from the mental fatigue of work or studies, contributing to improved work performance and satisfaction (Kaplan, 1995).

Why have an urban forest master plan?

- To guide the City's decision-making about trees in the city:
 - On the city's land
 - On other people's land
 - To help both the City and its citizens understand their urban forest better, in hopes of encouraging both to find ways to improve it and capture even more benefits from it over time
- 

City decision-making . . .

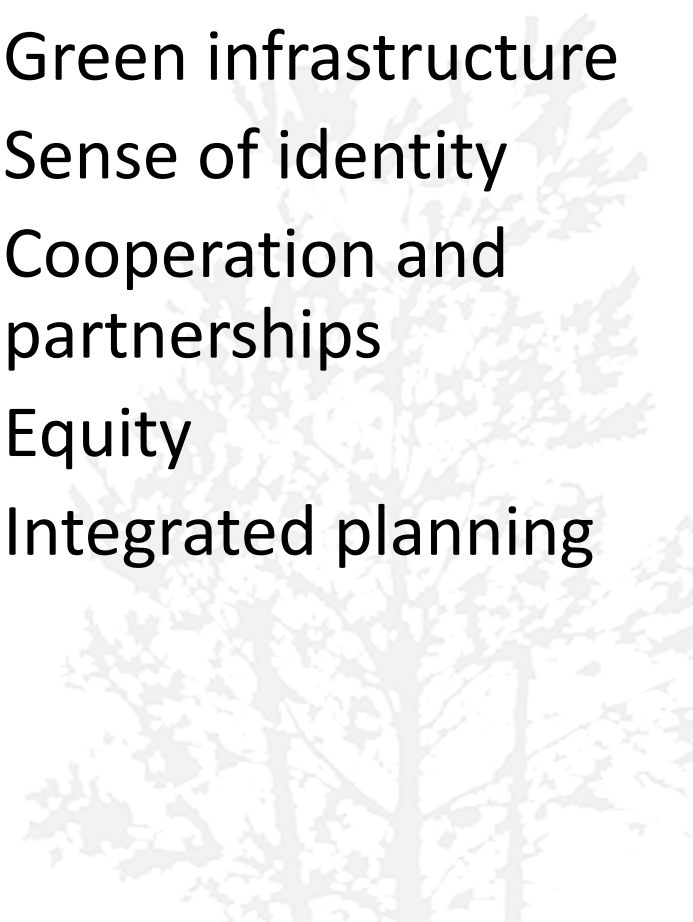
- On city land:
 - Planting – how many trees, which kinds, where
 - Maintaining – how, how often
 - Conflicts with infrastructure – power lines, new construction
- On other people's land:
 - Stewardship – advice, assistance, incentives,
 - Control – regulations for removal, retention, replacement

Guiding Principles of UFMP

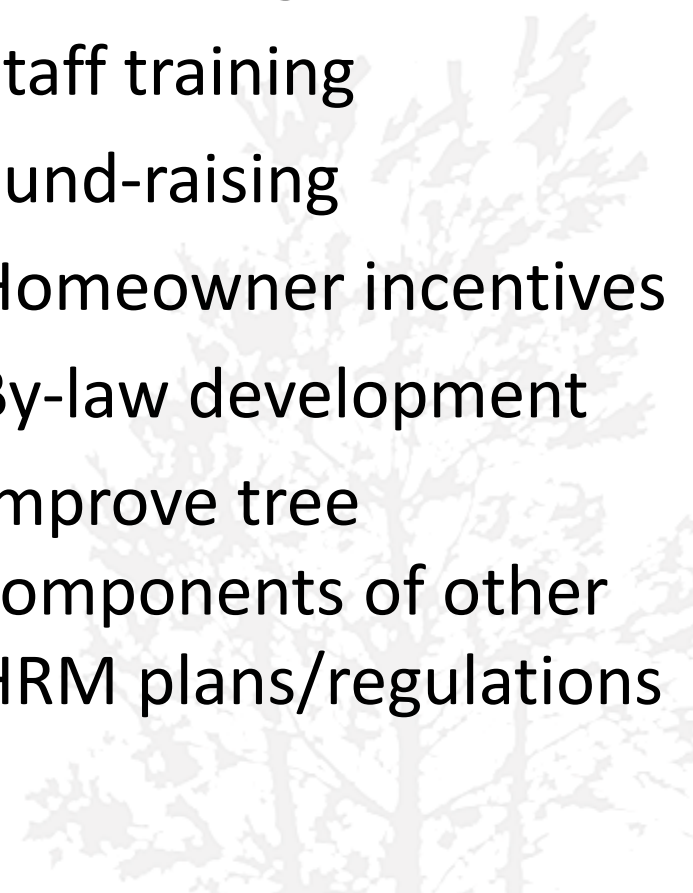
- Adaptive Management
- Precautionary Principle – “proceed with caution”
- Public Participation
- Sustainable Development



Operational Principles

1. Comprehensive approach
 2. Urban forest values
 3. Space and location
 4. Time and timing
 5. Climate change
 6. Naturalness
 7. Invasive species
 8. Public understanding
 9. Stewardship
 10. Priority-setting
 11. Green infrastructure
 12. Sense of identity
 13. Cooperation and partnerships
 14. Equity
 15. Integrated planning
- 

Actions needed (32 categories)

- Plant trees (neighbourhood treatments)
 - Maintain trees (7-yr pruning cycle)
 - Landowner/public education/programming
 - Support citizen-led stewardship initiatives
 - Research and monitoring
 - Staff training
 - Fund-raising
 - Homeowner incentives
 - By-law development
 - Improve tree components of other HRM plans/regulations
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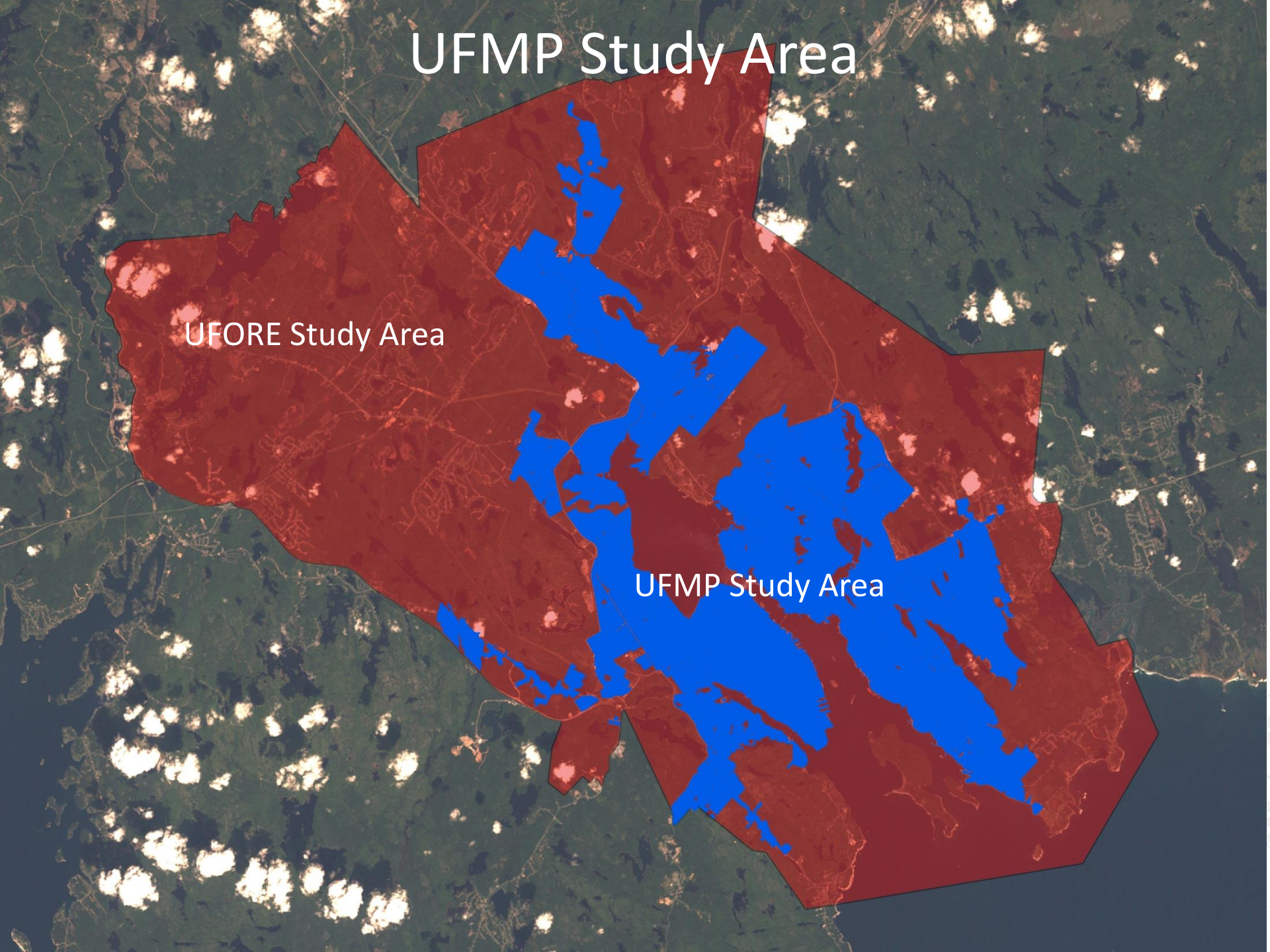
Four Levels of Urban-Forest Management

- ➔ **UFMP Study Area:** Ecological, economic, social values/objectives/indicators/targets, operational principles, and actions
- ➔ **Community:** Natural and human history, treatment summaries, and landscape patterns
- ➔ **Neighbourhood:** Local conditions, technical neighbourhood treatments, canopy targets
- ➔ **Neighbourhood Division:** Canopy targets, prioritization, collaboration

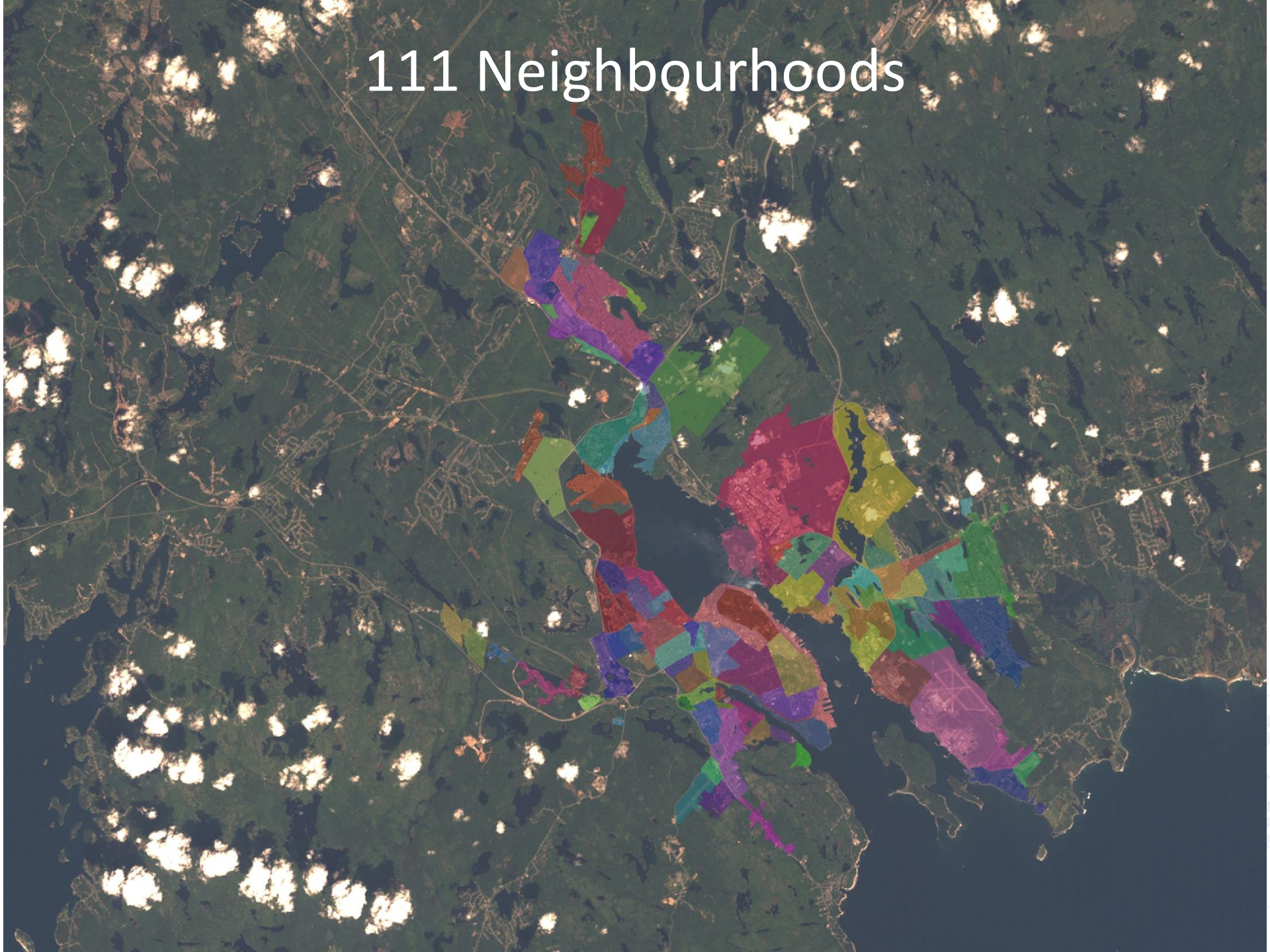
UFMP Study Area

UFORE Study Area

UFMP Study Area

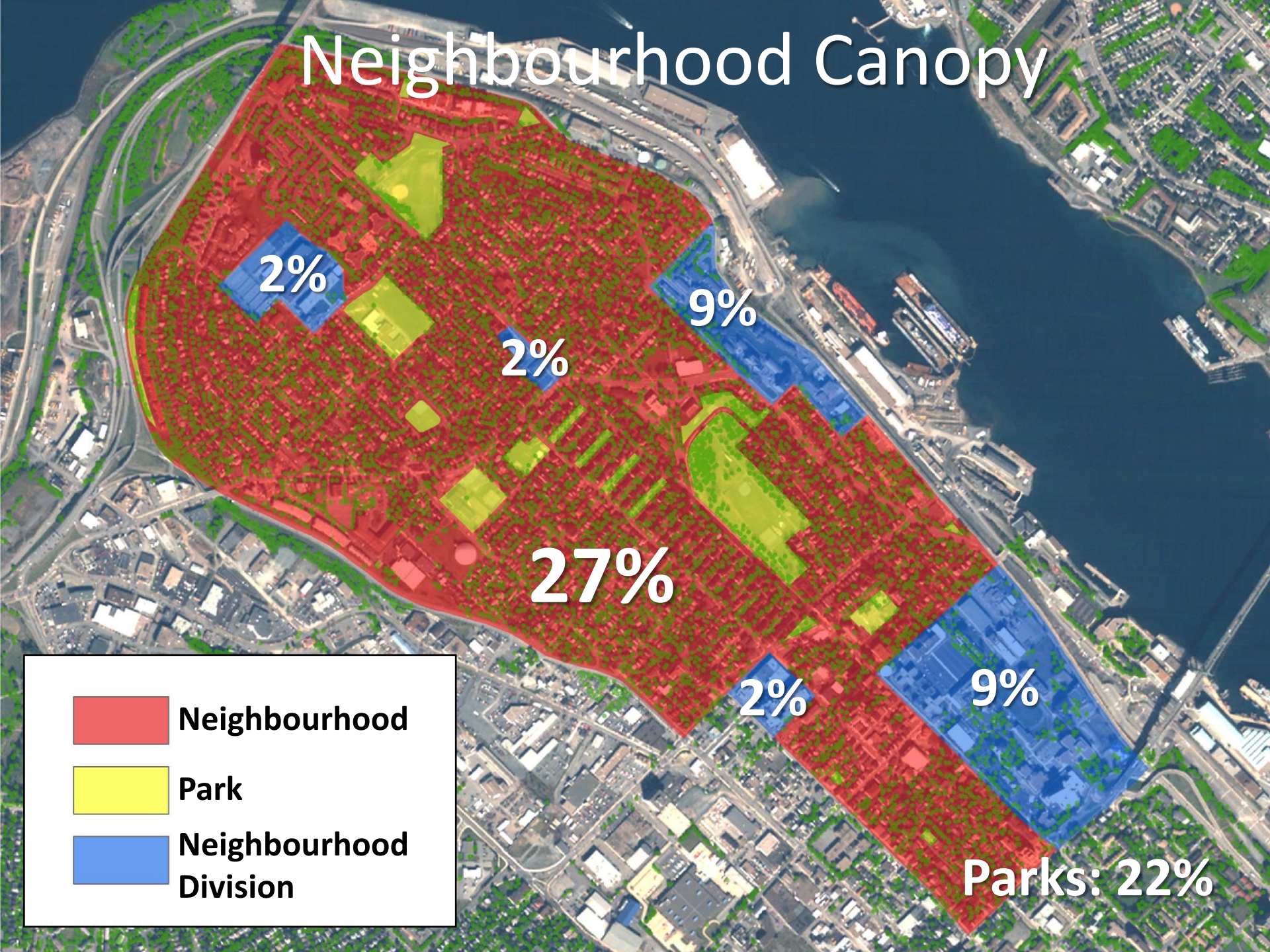
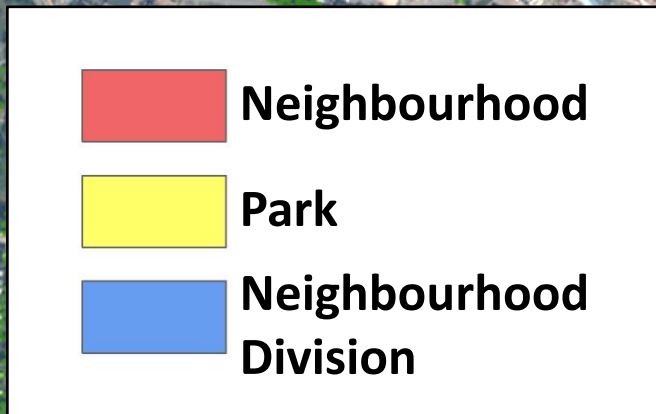


111 Neighbourhoods

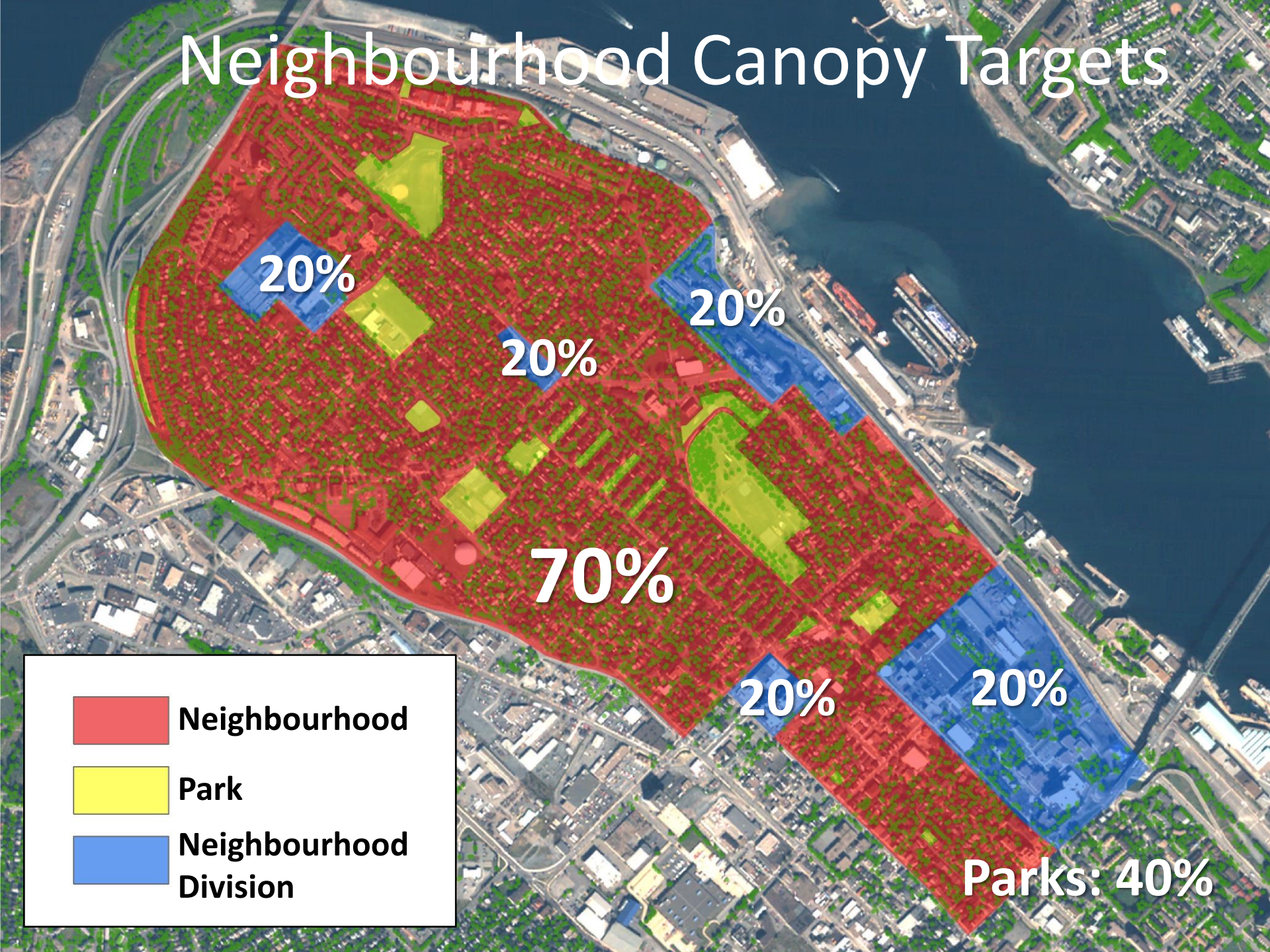
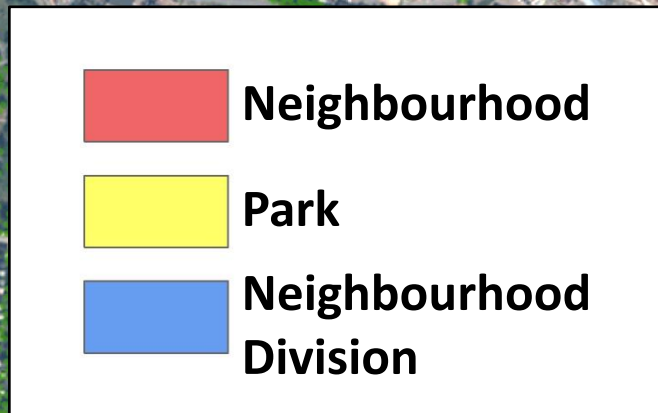




Neighbourhood Canopy



Neighbourhood Canopy Targets



North End

The North End urban forest neighbourhood (Figure 90) is bounded by Robie Street, Africville Memorial Park, Barrington Street, North Street, and Agricola Street. Throughout the Nineteenth and early Twentieth centuries, this neighbourhood was generally known as Richmond. It was a prosperous and attractive working-class district of Halifax until its destruction in the 1917 Halifax Explosion. The explosion devastated this and other areas of Halifax, killing 1,600 people and injuring 9,000 others. The pressure wave from the blast levelled most of the neighbourhood's urban forest. The subsequent restoration of this neighbourhood was led by Thomas Adams, a prominent town planner and early advocate of the "Garden City" movement, which favoured the inclusion of parks, trees, and open space in the design of new communities. Today's urban forest in the neighbourhood can be traced back to Adams' planning wisdom and insistence that intensive tree planting be part of the area's overall redevelopment. As new developments take place in this neighbourhood, special care should be taken to maintain this vision. Adams' influence is best seen in the treed boulevards of the Hydrostone district west of Fort Needham Memorial Park. It is also visible in the neighbourhood's numerous parks and the evenly aged, older canopy provided by street trees throughout the area. The North End has 27% overall canopy cover (shown in red). The area's parks (shown in green) have 22% canopy cover. Institutional and commercial divisions (shown in blue) have lower canopy cover; they include Stadacona (1) and Mulgrave Park (2) with 9% canopy cover, and the Nova Scotia Community College (3), Oland Brewery (4), and Novalea Centre (5) with 2% canopy cover. A belt pattern of urban forest around the northern curve of the neighbourhood is shaped by Africville Memorial Park and adjacent green spaces.

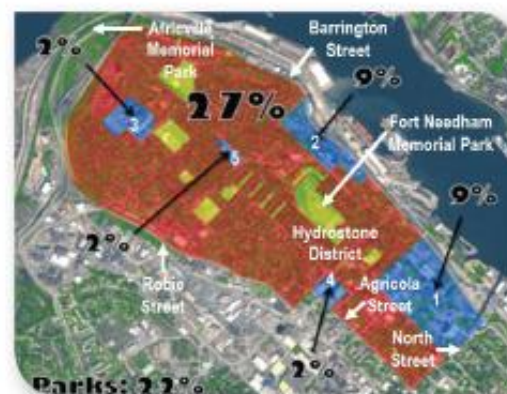
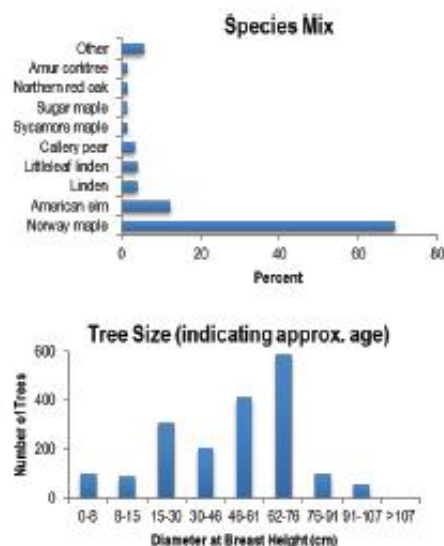


Figure 90

Challenges

- The North End's urban forest is aging, with an uneven distribution of tree sizes and a high proportion of older, larger trees.
- The North End's species diversity is low, with Norway maple and its cultivars constituting 65% of the population. Moreover, the representation of native species, particularly of Acadian old-growth species, is minimal.
- The institutional and commercial neighbourhood divisions (shown in blue in Figure 90) have marginal canopy cover. High levels of imperviousness and a lack of plantable sites are considerable challenges to improving canopy cover in these areas.

Opportunities

- A planned planting program will produce a sustainable age and species distribution of trees.
- Cooperative programs with institutional and commercial landowners could dramatically increase their share of the neighbourhood's canopy cover.
- There are 998 plantable sites for new street trees in this neighbourhood.
- Tree canopy in parks can be increased from 22% to 40%.
- Land use policies and land use by-laws will be amended to ensure that there are improvements to canopy cover when new developments take place.
- Local groups will be organized to work with HRM to achieve urban forest targets.

Neighbourhood Indicators - Actual and Targets

Indicator	Actual	Target
North End percent canopy cover	27	≥ 70
Stadacona percent canopy cover	9	≥ 20
Mulgrave Park percent canopy cover	9	≥ 20
Novalea Centre percent canopy cover	2	≥ 20
Nova Scotia Community College percent canopy cover	2	≥ 20
Oland Brewery percent canopy cover	2	≥ 20
Parks percent canopy cover	22	≥ 40
Percent of neighbourhood in forest stands	0.7	2.7
Percent of trees within the smallest size class	6	≥ 10
Percent of street trees that are native	15	50
Number of species that comprise more than 5% of all street trees	2 (Norway maple, American elm)	0
Number of genera that comprise more than 10% of all street trees	2 (Maple, elm)	0
Number of Acadian old-growth species that represent at least 1% of all street trees	2 (Sugar maple, red oak)	6 (red spruce, hemlock, white pine, sugar maple, yellow birch, red oak)
Percent of street trees with no structural problems	94	100
Number of plantable sites in HRM rights of way	998	0
Percent canopy cover of riparian and recharge zones	N/A	100
Percent of public play areas that are adequately shaded (40% canopy cover)	33	100
Percent of residential properties with at least one value tree	67	100

Biophysical Conditions

	Area (ha)	Impervious (%)	Surficial Geology
The North End	239	54	Till veneer, drumlins
Stadacona	21	64	Till veneer
Mulgrave Park	8	64	Till veneer
Novalea Centre	1	96	Till veneer
Nova Scotia Community College	5	58	Till veneer
Oland Brewery	2	100	Till veneer
Parks	16	19	Till veneer, drumlins

Socioeconomic Conditions

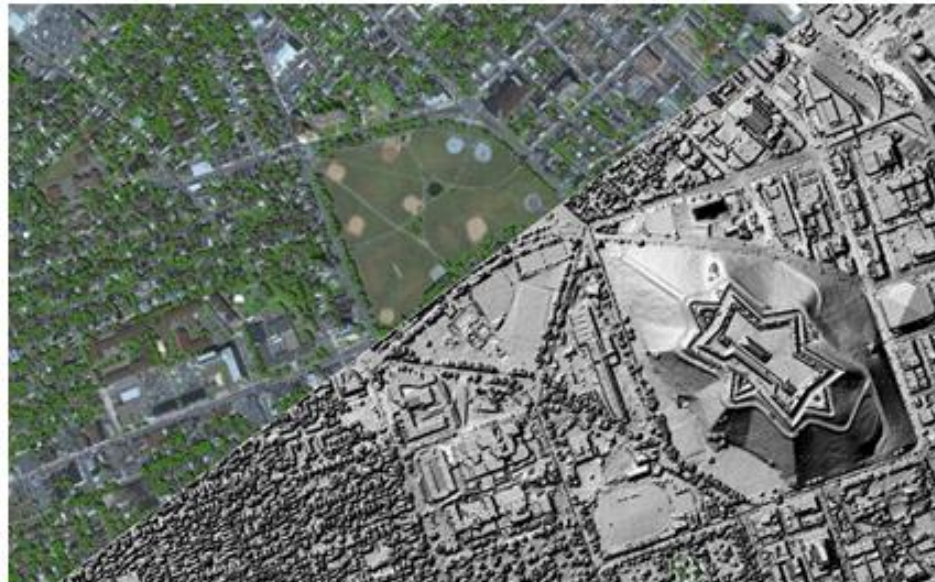
	Land Use	Zoning	Decade of Development
The North End	Residential	R-2, R-1, C-2	1920s
Stadacona	Institutional	CFB	
Mulgrave Park	Multi-family residential	R-3, C-2	1960s
Nova Scotia Community College	Institutional	R-2	1970s
Oland Brewery	Commercial	C-3	1930s
Novalea Centre	Commercial	C-2A	1960s
Parks	Park	P	1920s

Neighbourhood Treatments

Treatment	Prescription
Average number of trees to plant on HRM lands per decade	400
Area of naturalized forest stands to be created on HRM lands per decade	3 ha
Average number of native trees to plant on HRM lands per decade	260
Average number of Acadian old-growth species to plant on HRM lands per decade	8 (red spruce, eastern hemlock, white pine, yellow birch)
Refrain from planting the following genera until 2020	Maple, elm
Average number of trees with structural problems to remove or maintain, unless otherwise required for safety reasons	45
Additional Neighbourhood Treatments	
Target the rich drumlinoid soils of the Fort Needham Memorial Park area for plantings of naturalized stands and native Acadian species.	
Establish collaborative tree-planting efforts with representatives of the institutional and commercial divisions.	

Urban Forest Canopy Data



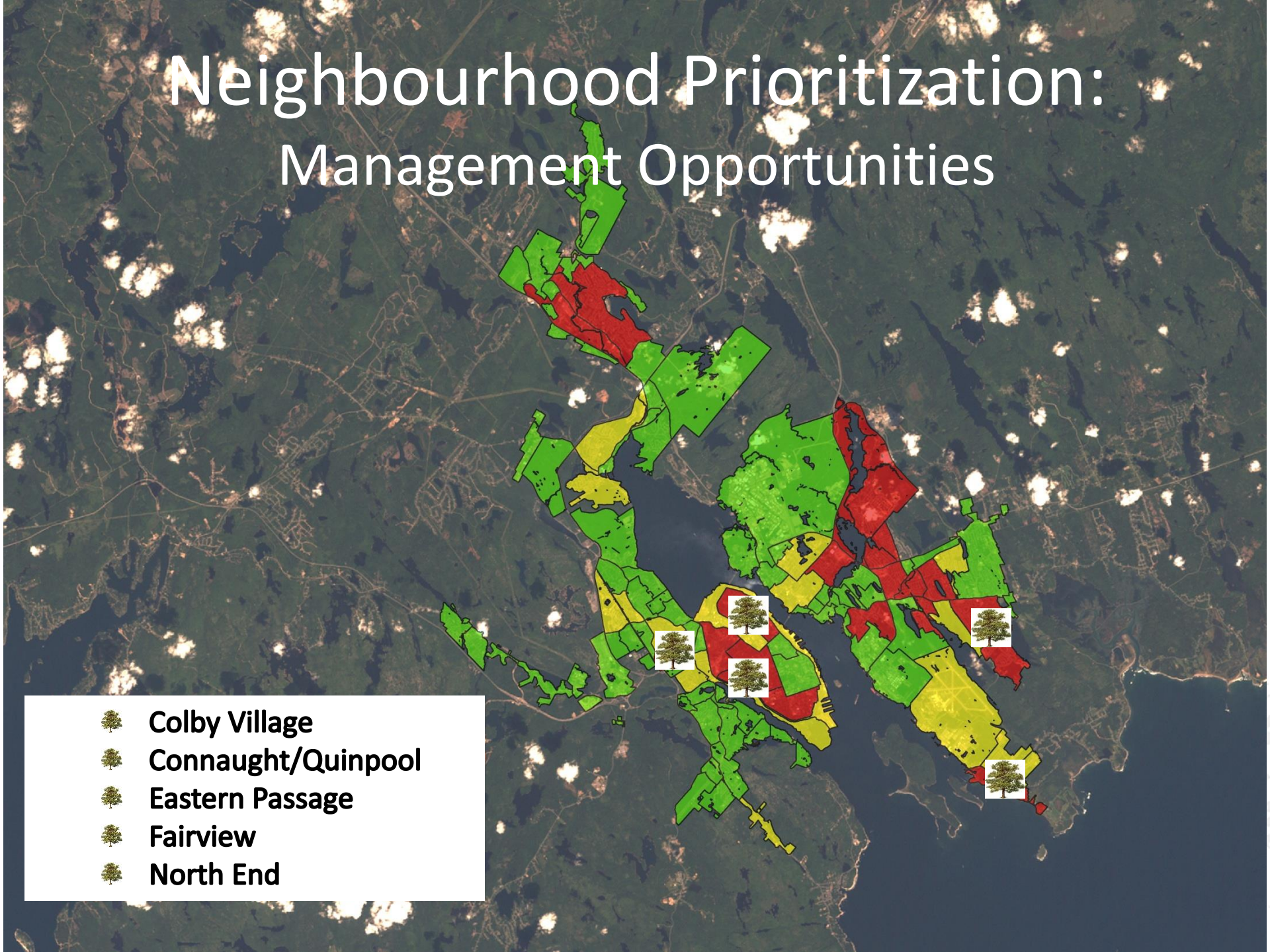


**Development of an Urban Forest Canopy Model for input into a Lidar-based
Stormwater Runoff Model for Halifax Harbour Watersheds**

Suzanne Monette and Christopher Hopkinson, Applied Geomatics Research Group (AGRG) -2010
Nova Scotia Community College
Annapolis Valley Campus
50 Elliott Road
Lawrencetown, NS BOS 1M0

May 2012

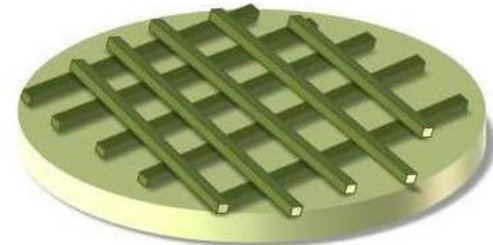
Neighbourhood Prioritization: Management Opportunities

- 
- The map shows a coastal region with various land parcels. A large area in the center is colored green, with several smaller green patches scattered around. A prominent red area is located in the upper left. Yellow areas are found in the center and lower right. Five tree icons are placed on the map: one in the lower left green area, one in the central red area, one in the central yellow area, one in the lower right red area, and one in the far right yellow area.
- Colby Village
 - Connaught/Quinpool
 - Eastern Passage
 - Fairview
 - North End

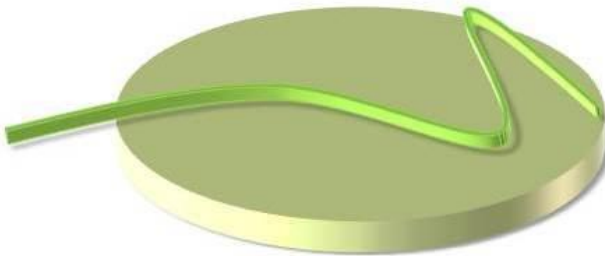
Landscape Patterns



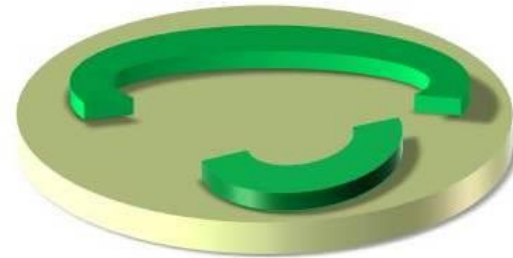
Scattered: the patchwork



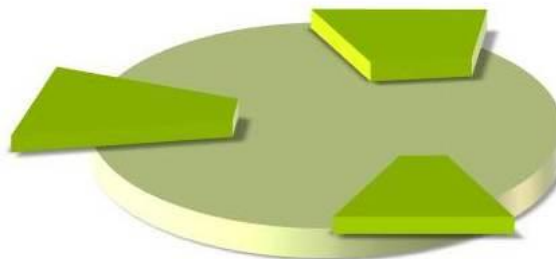
Linear: the network



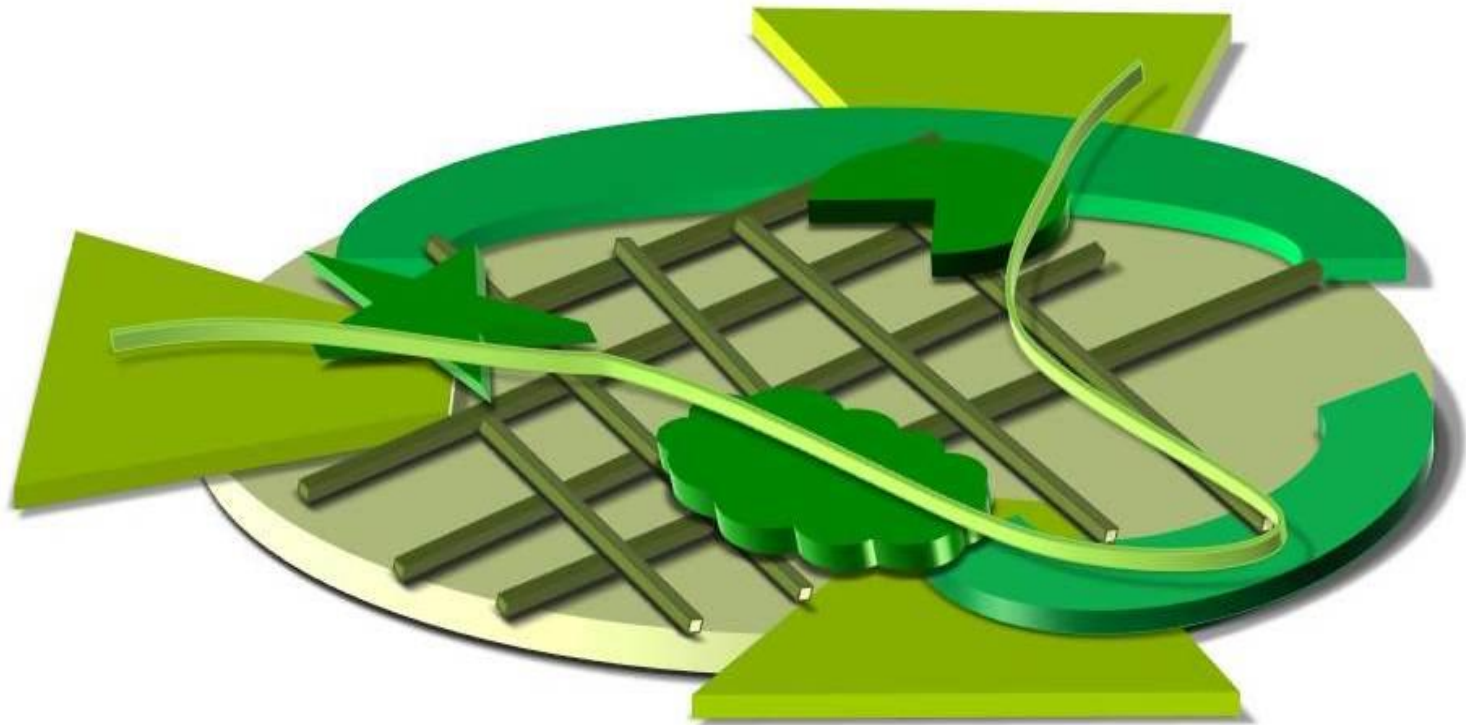
Linear: the ribbon



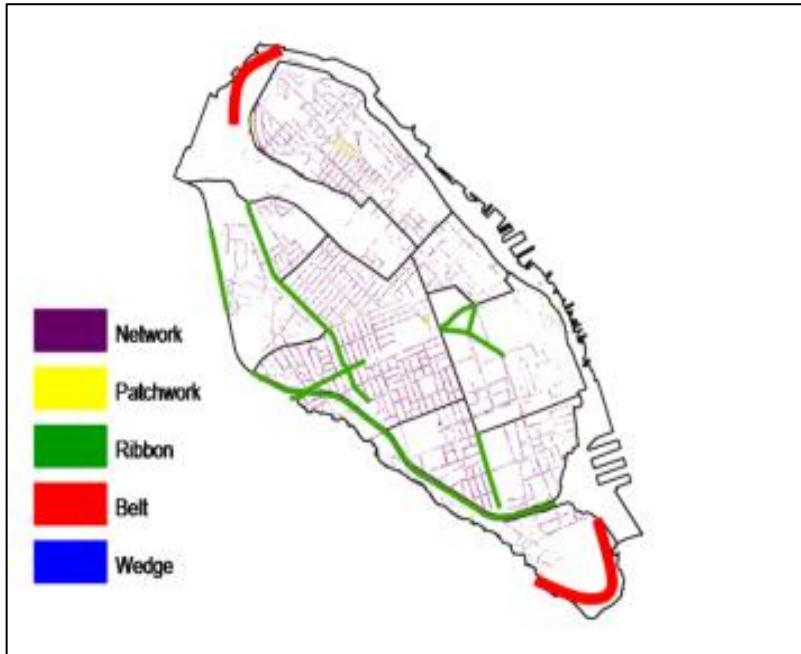
Area: the belt



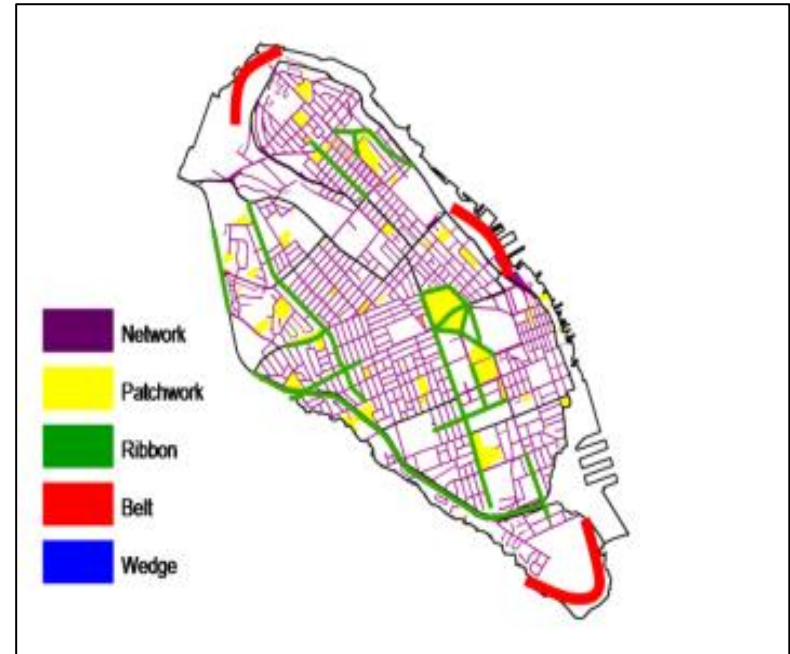
Landscape Patterns



Community Landscape Patterns

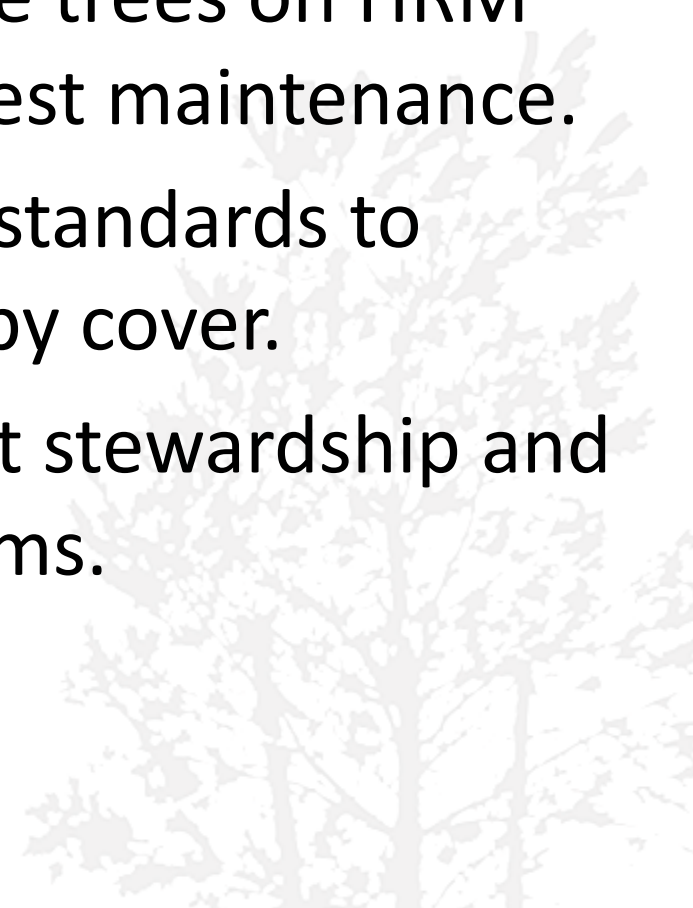


Halifax Peninsula Today



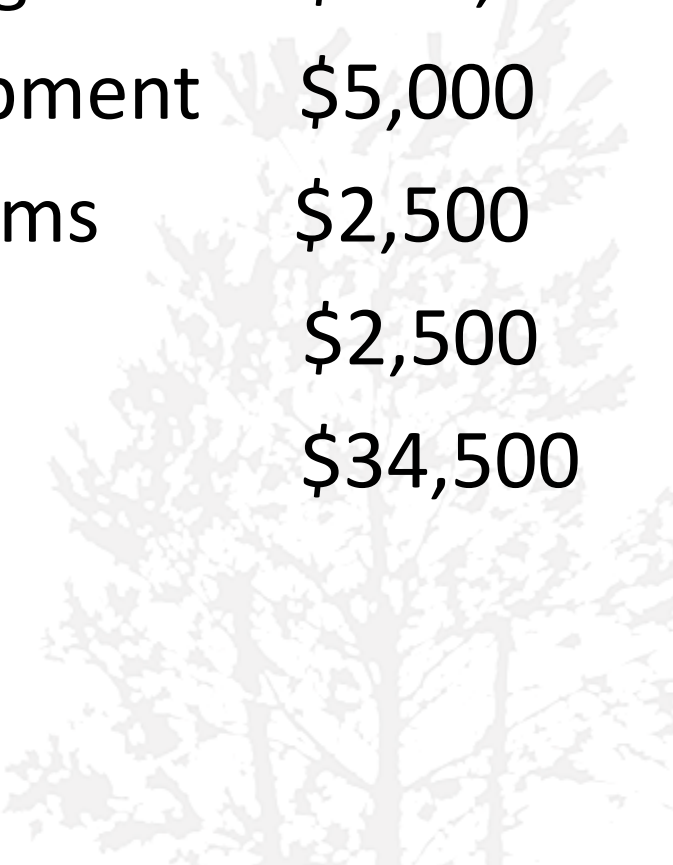
Halifax Peninsula Tomorrow

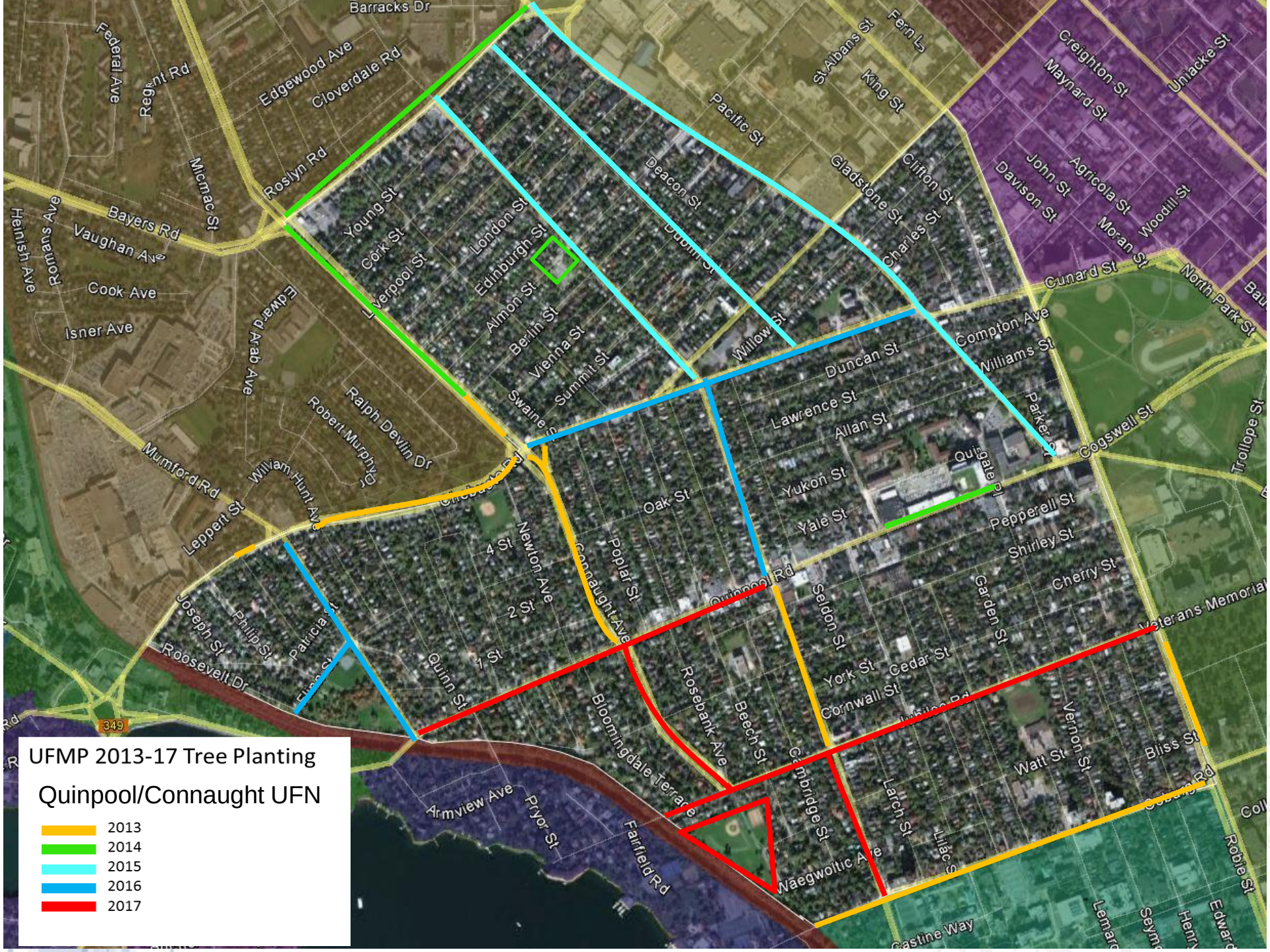
Implementation Priorities

1. Increase funding, plant more trees on HRM land and improve urban forest maintenance.
 2. Adopt new regulations and standards to conserve urban forest canopy cover.
 3. Promote citizen urban forest stewardship and develop educational programs.
- 

Implementation Strategy 2016/17

Cost Estimates

- 
- | | |
|----------------------------------|-----------|
| 1. Street tree planting | \$650,000 |
| 2. Cyclical street tree pruning | \$350,000 |
| 3. By-law/regulation development | \$5,000 |
| 4. Public stewardship programs | \$2,500 |
| 5. Urban orchard expansion | \$2,500 |
| 6. Research and monitoring | \$34,500 |



UFMP 2013-17 Tree Planting
Quinpool/Connaught UFN

- 2013
- 2014
- 2015
- 2016
- 2017



UFMP 2013-17 Tree Planting
Fairview UFN

- 2013
- 2014
- 2015
- 2016
- 2017



UFMP 2013-17 Tree Planting
North End UFN

- 2013
- 2014
- 2015
- 2016
- 2017

Cyclical Street Tree Pruning

- 7 year pruning cycle target
- Maintain reactive pruning
- Implement proactive pruning
- Renew MOU for HRM/NSPI Vegetation Management Program
- Develop NSPI cost-share agreement and pruning protocols:
 - promoting UFMP tree health and canopy objectives
 - providing for safe and reliable operation of NSPI's distribution system

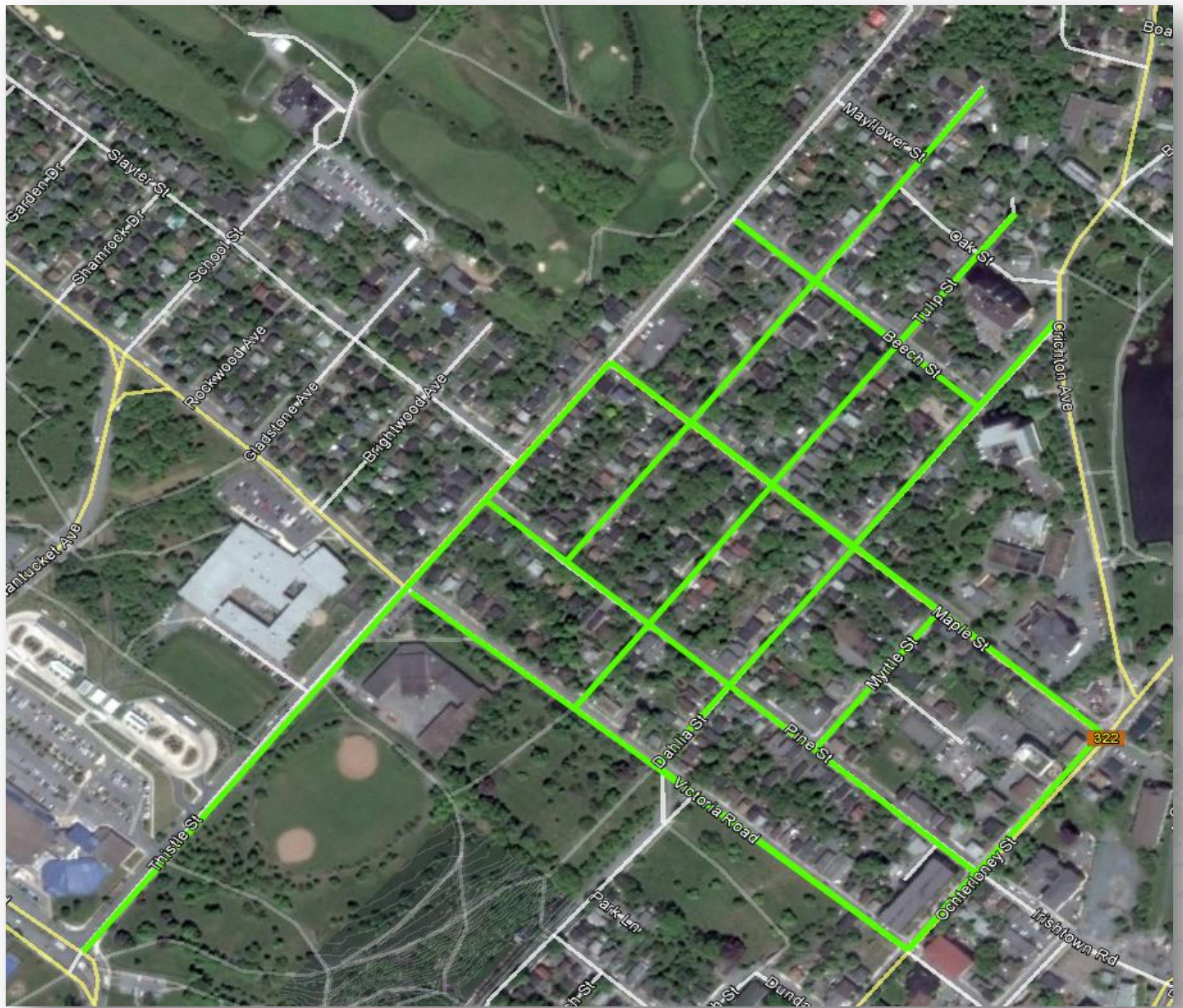


Urban Forest Master Plan (UFMP) 2015 Cyclical Tree Pruning Project

Tentative start date: Wednesday, January 14

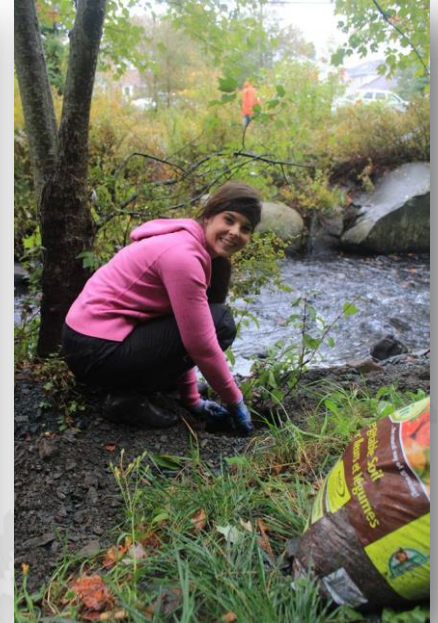
Estimated completion date: Friday, February 13

Work Location (Street)	Between	And	Comments
Thistle St.	Wyse Rd.	Maple St.	Proposed start date: Wednesday, Feb. 2 Estimated completion date: Friday, Feb. 6
Rose St.	Pine St.	Oak St.	Proposed start date: Wednesday, Jan. 14 Estimated completion date: Friday, Jan. 16
Tulip St.	Victoria Rd.	Oak St.	Proposed start date: Monday, Jan. 19 Estimated completion date: Friday, Jan. 23
Dahlia St.	Victoria Rd.	Beech St.	Proposed start date: Monday, Jan. 26 Estimated completion date: Friday, Jan. 30
Myrtle St.	Pine St.	Maple St.	Proposed start date: Monday, Jan. 26 Estimated completion date: Friday, Jan. 30
Ochterloney St.	Victoria Rd.	Maple St.	Proposed start date: Monday, Jan. 26 Estimated completion date: Friday, Jan. 30
Victoria Rd.	Thistle St.	Ochterloney St.	Proposed start date: Monday, Feb. 2 Estimated completion date: Friday, Feb. 6
Pine St.	Thistle St.	Ochterloney St.	Proposed start date: Monday, Feb. 2 Estimated completion date: Friday, Feb. 6
Maple St.	Thistle St.	Ochterloney St.	Proposed start date: Monday, Feb/ 9 Estimated completion date: Friday, Feb. 13
Beech St.	Thistle St.	Dahlia St.	Proposed start date: Monday, Feb. 9 Estimated completion date: Friday, Feb. 13
			NOTE: Proposed start dates and estimated completion dates are weather dependent.



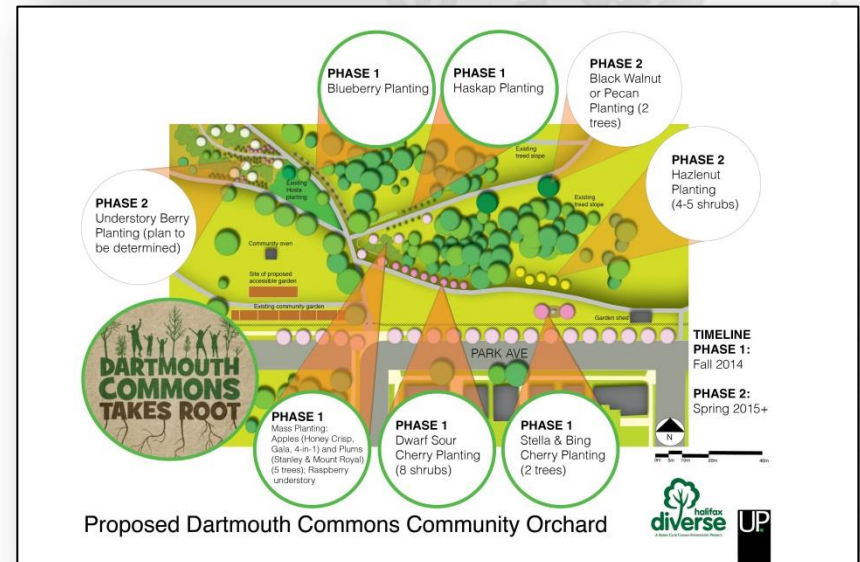
Volunteers & Public Education

- 33 hectare annual planting target
- Focus on naturalization
- Hardwood whips
- Coniferous seedlings
- 600 volunteers in 2015
- 7000+ trees planted



Volunteers & Public Education

- Public engagement
- ENGO partnerships
- Program development
- Pilot programs
- Volunteer recruitment



Regulations -2016

- Dalhousie Planning School Research
- Riparian protection by-law development
- Canopy protection by-law development
- Tree protection standards - infrastructure

LUC NEWS

Councillors 'bark' at each other over tree by-law

By ISAAC OLSON

An axe is one of the best tools for splitting wood and, as it turns out, a municipal tree by-law is one of the best ways to split Côte St. Luc's council — a council that, for the most part, doesn't see too many disagreements as it lumbers through the usual round of monthly votes.

While there are those that stand by the approximately two-year-old by-law's overall design — a design that has the council voting on every single request to fell a private tree — there are others that say it is a hatchet job that is causing discord among elected officials. Still, at the root of the debate, are those councillors who sit on the fence and "yea" or "nay" each vote on a case-by-case basis. Now the discussion

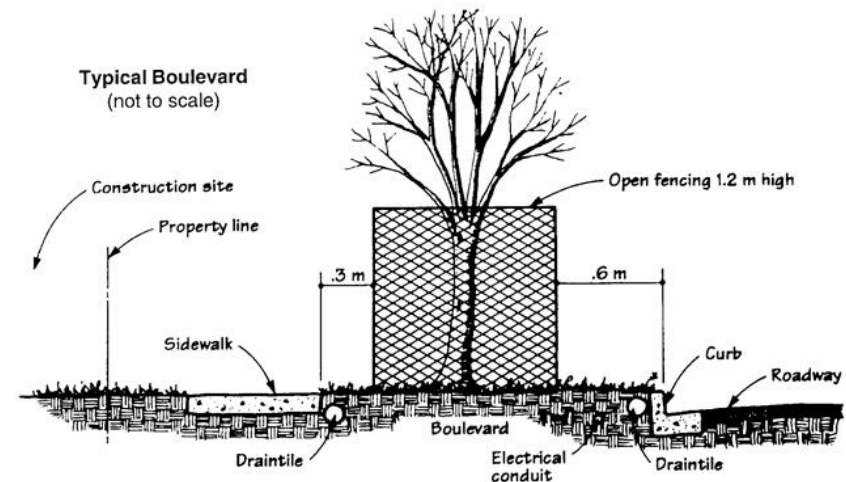
voted off the chopping block at another.

Her solution is to have staff, namely the horticulturist and urban planning director, apply the by-law "fairly and equitably to all homeowners."

Kovac: Give owners more say

"When the by-law was first enacted, the vote was six to two and I predicted it would certainly evolve, change and cause all these problems because, at some point, it does become the resident's right to enjoy their property," began Councillor Ruth Kovac.

"I also think it becomes prohibitively expensive for some people when they have to fell a large tree for whatever reason and then they are obliged to have more than one tree to replace it can become very, very costly."



Current Opportunities

- Integrate UFMP policies with:
 - Centre Plan
 - Greenbelting and Public Spaces Priorities Plan
 - Subdivision By-law amendments
- Include UFMP implementation strategy in 2016/17 HRM business plans.

Monitoring

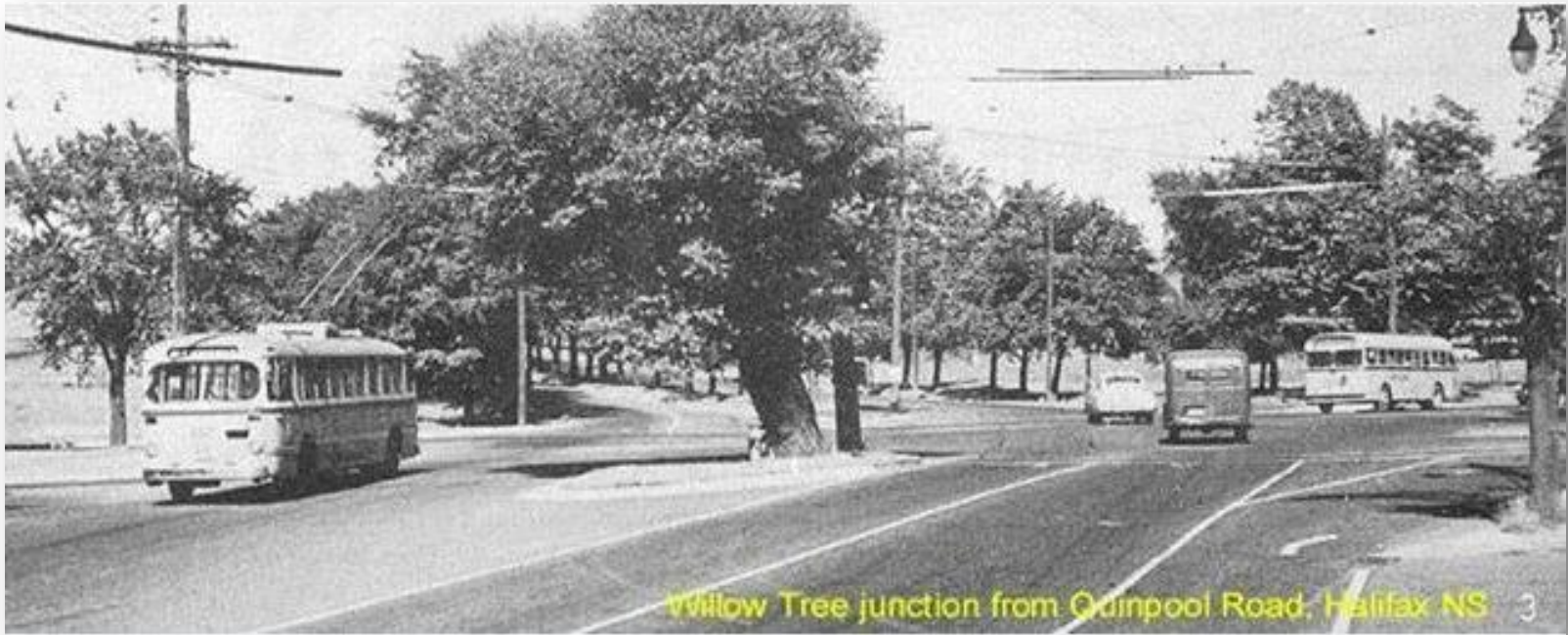
Variable name:	Age class diversity
Relates to an action, value or principle:	A2. Apply neighbourhood planting treatments for street trees. V2. Species diversity and age-class diversity
Action or system-condition/outcome:	System-condition
Units of measurement:	Diversity to be determined through a comparison of the percentage of trees in each age class. Age of individual trees in years – calculated through approximation based on tree size characteristics (dbh, height, and crown diameter)
Monitoring method:	Tree height, dbh, and crown diameter can be determined through field sampling, LiDAR, aerial photography, and/or satellite imagery. UFORE/i-Tree data collection protocols. In the previous tree inventory diameter at breast height (dbh) was measured and used in the UFMP as a surrogate for age. However, true ages for trees planted by HRM will be known from the planting records.
Timing specifics:	The Plan recommends a UFORE/i-Tree assessment 5 years after approval and again after 10 years. The data should be available before the scheduled reviews of the Plan.
Spatial specifics:	The new assessment should at minimum cover the same area as the previous survey and at best should also include new areas of development that would be considered part of the serviced core of HRM and areas slated for development within the next 5 years.
Additional notes:	
References:	Powell, D. (2005). How to measure a big tree. USDA. Retrieved from http://www.fs.usda.gov/Internet/FSE_DOCUMENTS/stelprdb5_202838.pdf Imai, Y., Setojima, M., Yamagishi, Y., & Fujiwara, N. (2004). Tree-height measuring characteristics of urban forests by lidar data different in resolution. <i>International Archives of Photogrammetry Remote Sensing and Spatial Information Sciences</i> , 35, 513-5. Koukoulas, S., & Blackburn, G. A. (2005). Mapping individual tree location, height and species in broadleaved deciduous forest using airborne LIDAR and multi-spectral remotely sensed data. <i>International Journal of Remote Sensing</i> , 26(3), 431-455.

- Dalhousie SRES research partnership
- Continue guidelines development
- Initiate transition to iTree indicator measurement
- Inventory and monitor 2013-2016 planting and pruning

Asset Management



Replant the “Willow Tree”*



* *Never, never, never give up.*
Winston Churchill



Private citizen Bill Mont replaced this Willow Tree, victim of a car crash.

